Sprint 2 Review

Advisor: Dr. Pooja Viswanathan

Feedback - Areas to Improve from Sprint 1

Area Improve 1:

- Advisor comment: Work towards creating an actual MVP that focuses on end-to-end user experience and not just end-to-end functionality on the features independently.
- Why this matters: The scope of our sprint explicitly called for progress toward an MVP. However, in earlier iterations we primarily developed features in isolation, which meant we did not yet have a cohesive product that a user could meaningfully interact with from start to finish.
- Sprint 2 changes made: During Sprint 1, we focused almost entirely on backend functionality and had little to no user experience or front-end integration. In Sprint 2, we prioritized connecting the entire pipeline. From user input to system output, we were able to establish a coherent workflow. This allowed us to validate that core components work together as a unified system and to begin shaping a usable interface that reflects how a real user would interact with the product.

Area Improve 2:

- Advisor comment: Investigate alternative, more accurate approaches to 3D reconstruction beyond the use of ArUco markers.
- Why this matters: The ArUco-based approach produced limited accuracy and imposed significant setup overhead, as markers needed to be placed on many surfaces throughout the room. This made the system less scalable, more time-consuming to deploy, and less practical for real-world use.
- Sprint 2 changes made: During Sprint 2, we researched and evaluated markerless 3D reconstruction methods, including COLMAP, which uses structure-from-motion and multi-view stereo techniques to generate higher-fidelity reconstructions from images alone. This investigation informed our plan to transition toward a more robust and scalable reconstruction pipeline in future iterations.

Peer Feedback on Sprint 1 Demo

Disclaimer: Some of our comments are repeating/don't contain any constructive feedback for future implementations, so they are not included in this document. Some other comments include functionalities that were already planned (such as accessible UI and compliance suggestions) so they were also not included in this document.

Area Improve 1:

Zaki Hasan Ali

January 6 at 5:44 PM

Great presentation! The live demo of the application working in a bathroom and detailed analysis over the code showing how the logic works was done very well. Additionally, being in contact with an individual in a wheelchair is also a good step to be able to capture the use case in the best way possible. Have you considered expanding this to other applications as well such as buildings as a whole rather than just bathrooms? This auditing system could be very useful for auditing entrances and hallways and other areas in a building for accessible people as well.

Consideration of peer feedback: Expand this project to other applications such as buildings

- Consideration status: Deferred.
- Explanation: Although expanding this application is a great suggestion, it is not feasible in the time frame that we've been given. Our focus is on auditing washrooms specifically for this capstone project.

Area Improve 2:

Feedback

Good presentation! I like the idea! I think your team choose the right option to focus more on the backend in sprint 1 this being the hardest part of your project. In sprint 2 make sure to focus heavily on the UI because this software is very useful but without proper UI could hinder it's usefulness.

Some ideas/ suggestions I have

- I know you have it right now to give a scale based on the objected in a room and return a colour coordinated score. Could it be use full to give a documented suggestion list? First stating why (which I believe will be your audit) but then you could make some suggestions to the user using the accessibility standards. ex. "If the handle bar was placed 1 inch higher it would be more in compliance because the average height of a human ... and there center of gravity ..."
- I am not sure how accurate the computer vision is on the model you are using but it could be useful to include a error percentage for cases that the model detected could be out of scale, etc.
- If you guys do decide to make the app a public use case where people who have accessibility needs, then it could be useful for them to have some form where they could notify the washroom they are in and say why or why not it is accessible to make this a high priority audit location.

Consideration of peer feedback: Include error percentage for cases

- Consideration status: Partially considered
- Explanation: This was definitely a consideration when we used AruCo markers primarily to reconstruct the 3D space. However, after switching to COLMAP, the error percentage is consistently lower than 1%. Due to the negligible error, we ended up not implementing this consideration.

Consideration of peer feedback: If the app is a public use case, add a form for users so they can specify the washroom location and why it should be high priority

- Consideration status: Deferred
- Explanation: Although this is great advice, it's specific to if the app is a public use case, which in this project scope it is not.

Area Improve 3:

Michael George Brian Botelho

January 8 at 9:43 AM

Great job on sprint 1! The 3D models look great so far and I am excited to see how you integrate it in your webapp. One thing I think could be good to add would be some way for users to re-scan a bathroom or even mark them as non-compliant if something was changed that made it harder to use that space. That could help ensure the scans and information in your app is as up-to-date as possible.

Consideration of peer feedback: Allow users to re-scan a bathroom or mark them as non-compliant

- Consideration status: Deferred
- Explanation: The users are able to rescan any washroom at any given point, and our backend pipeline will mark it as non-compliant if that space has become inaccessible. This is already implemented in our project.